

Population Aging, Social Security Spending, and Informality:

The Financing Margin in Colombia

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Abstract

This paper studies how population aging affects capital accumulation and labor informality when the government finances a fixed social-security outlay per older adult. Let $d_t = N_t^o/N_t^y$ denote the ratio of people aged 65 and over to the working-age population. Government outlays per working-age adult obey the identity $G_t^{SS} = \bar{g}_o d_t$. I embed this fiscal block in an annual deterministic general-equilibrium model in which private capital is used only in formal production and workers choose between formal and informal employment. The Colombian illustration raises d_t from 15.02 percent in 2024 to 27.28 percent in 2050, using DANE population projections, while holding $\bar{g}_o$ fixed. Initial public pension outlays are normalized to 5.7 percent of GDP. If broad taxes finance the entire aging-induced increase, stationary capital and informality do not change and private consumption falls 5.8 percent. If half of the increase is charged to the formal wage bill, capital falls 13.2 percent and informality rises 5.9 percentage points. If that share is 75 percent, capital falls -22.9% and informality rises 10.2 percentage points. Financing the entire increase on formal payroll has no interior stationary equilibrium in this calibration. The model treats social-security outlays as a real resource claim. It is therefore a reduced-form aging exercise, not a literal model of pension transfers; that interpretation requires separate generations and retired-household budgets.

Keywords: population aging; social security; informality; capital accumulation; payroll taxation; Colombia. **JEL:** E22; E62; H55; J11; J46.

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1 Introduction

Population aging creates a fiscal pressure even when policy rules do not change. If every older adult receives a fixed real amount of pensions, health care, or long-term care, total social-security outlays rise with the number of older people relative to the tax-paying population. The relevant primitive is therefore not an arbitrary path for government purchases. It is the demographic ratio that maps the number of beneficiaries into a fiscal claim.

This paper studies that mechanism in Colombia. Let N_t^o be the population aged 65 and over and N_t^y the population aged 15–64. The government assigns a fixed real outlay $\bar{g}_o$ to each older adult. After normalizing the model by the working-age population, social-security spending is

$$G_t^{SS} = \bar{g}_o \frac{N_t^o}{N_t^y} = \bar{g}_o d_t.$$

DANE projections imply that d_t rises from 15.02 percent in 2024 to 27.28 percent in 2050 ([Departamento Administrativo Nacional de Estadística, 2025b](#)). Holding $\bar{g}_o$ fixed converts that demographic change into a permanent increase in the government’s revenue requirement.

The financing rule determines how the demographic pressure reaches the rest of the economy. A tax on the formal wage bill lowers the relative return to formal employment and shrinks its own base. A sufficiently broad tax does not create that formal–informal wedge in the present model. Public debt could spread the adjustment over time, but it cannot finance a permanently larger old-age population without a corresponding long-run primary revenue stream.

The model is annual and deliberately parsimonious. A representative household chooses private consumption and saving. Formal firms use capital and labor, whereas informal production is labor intensive. Workers select formal or informal employment according to after-tax wages and heterogeneous sector preferences. A smooth discrete-choice margin keeps both sectors active. The capital stock follows the Euler equation and the aggregate resource constraint.

The initial calibration matches 55.4% national labor informality, measured by [Departamento Administrativo Nacional de Estadística \(2025a\)](#), and sets public pension outlays to 5.7% of GDP, the Colombian value reported by [Organisation for Economic Co-operation and Development \(2023\)](#). The demographic experiment keeps the outlay per older adult fixed and replaces the 2024 dependency ratio with DANE’s 2050

projection. The remaining parameters are provisional. The exercise is consequently a mechanism decomposition, not a forecast of Colombian pension or health expenditure.

The stationary results separate the resource claim from the formal-employment wedge. Under broad-tax financing, capital, output, and informality remain at their initial values; private consumption falls 5.8 percent. If half of the additional outlay is financed from the formal wage bill, informality rises 5.9 percentage points, capital falls 13.2 percent, and output falls 6.2 percent. When 75 percent is charged to formal payroll, informality rises 10.2 percentage points and capital falls 22.9 percent. A rule that places the entire increase on formal payroll does not admit an interior stationary equilibrium. That last result is an endogenous fiscal limit, not a numerical convergence failure.

One accounting qualification is essential. Cash pensions are transfers between households, not government absorption of output. Health and long-term care are closer to real public consumption. The current representative-household model treats the entire G_t^{SS} as a real resource claim. This shortcut permits a transparent analysis of capital and informality, but it cannot describe intergenerational redistribution or retired households' saving. A full pension interpretation requires an overlapping-generations extension.

The paper contributes a compact link between demography, fiscal financing, and informality. The next section places the exercise in the literature. The remaining sections present the model and calibration, report stationary and transition results, discuss the conceptual limits, and document the numerical algorithm.

2 Related literature

This paper connects three strands of work. The first studies the fiscal effects of population aging. For Colombia, [Ospina-Tejeiro et al. \(2024\)](#) project the public cost of Colpensiones and special regimes and show that longevity, growth, and formality materially change the long-run fiscal outlook. Their exercise models pension rules and coverage in detail. The present paper takes a narrower route: it fixes the outlay per older adult and isolates the general-equilibrium response of capital and informal employment.

The second strand concerns fiscal policy in general equilibrium. In the neoclassical model of [Baxter and King \(1993\)](#), persistent government purchases affect private wealth and investment. [Barro \(1990\)](#) distinguishes unproductive government consumption from productive public services, while [Galí et al. \(2007\)](#) shows that household heterogeneity and market structure can change private-consumption responses. Here the demographic path is exogenous, but its financing changes the formal labor wedge.

The third strand studies informality. [Levy \(2008\)](#) emphasizes that payroll-financed social benefits can tax formal employment and alter the allocation of labor. [La Porta and Shleifer \(2014\)](#) and [Ulyssea \(2018\)](#) show why informal firms cannot be represented only as tax-evading versions of formal firms. Productivity, technology, firm size, and selection differ across sectors. The current model retains only the labor-allocation margin and should be interpreted accordingly.

For Colombia, [Mejía and Posada \(2007\)](#) study formal and informal production when regulation can be evaded and enforcement is costly. [Osorio-Copete \(2016\)](#) examines tax reform and informality in dynamic general equilibrium. [Rincón et al. \(2017\)](#) embed government consumption, public investment, and a fiscal rule, while [Granda and Morales \(2023\)](#) analyze taxes, enforcement, and labor-market rigidities. [Herrera-Rojas et al. \(2025\)](#) provide a richer Colombian fiscal model with several revenue sources, expenditure categories, debt, and sovereign risk.

The contribution here is not comparable institutional detail. It is the identity $G_t^{SS} = \bar{g}_o d_t$ combined with an endogenous formal tax base. That structure makes visible which effects come from aging itself and which come from the instrument used to finance it.

3 Model

3.1 Household preferences and capital accumulation

Time is discrete and one period is one year. The representative household maximizes

$$U_0 = \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma} - 1}{1-\sigma} + \chi_G \log G_t^{SS} \right], \quad \beta \in (0, 1), \quad \sigma > 0. \quad (1)$$

Private consumption is C_t . The term G_t^{SS} denotes social-security outlays per working-age adult. This public term is optional and additive. Because it is exogenous to the household, it does not enter the first-order condition for private saving:

$$\frac{C_{t+1}}{C_t} = [\beta(1 + r_{t+1})]^{1/\sigma}. \quad (2)$$

Changing χ_G changes welfare evaluated along an allocation but does not change the allocation itself.

Private capital evolves according to

$$K_{t+1} = (1 - \delta)K_t + Y_t - C_t - G_t^{SS}. \quad (3)$$

Equation (3) treats social-security spending as a real resource claim. This is natural for publicly provided health and care services. It is a reduced-form shortcut for cash pensions, which are transfers rather than aggregate absorption. A literal pension model must assign benefits and taxes to different generations.

3.2 Formal and informal production

Formal firms use private capital and formal labor:

$$Y_t^F = K_t^\alpha (L_t^F)^{1-\alpha}, \quad \alpha \in (0, 1). \quad (4)$$

The competitive formal wage and net return on capital are

$$w_t^F = (1 - \alpha) \left(\frac{K_t}{L_t^F} \right)^\alpha, \quad (5)$$

$$r_t = \alpha \left(\frac{L_t^F}{K_t} \right)^{1-\alpha} - \delta. \quad (6)$$

Informal output uses labor only:

$$Y_t^I = A_I L_t^I, \quad w_t^I = A_I. \quad (7)$$

The labor force has unit mass, so $L_t^I = 1 - L_t^F$. Total output is

$$Y_t = Y_t^F + Y_t^I. \quad (8)$$

Equation (7) is a benchmark, not a claim that informal firms literally use no equipment. It represents labor-intensive services and microenterprises with limited access to formal finance. The assumption also implies that a shift out of formal employment reduces the labor input used together with private capital. A robustness exercise should allow informal firms to use a smaller amount of capital and should introduce a financing wedge rather than assume frictionless mobility of capital across sectors.

3.3 Endogenous informality

Workers differ in their idiosyncratic valuation of the two sectors. Aggregating extreme-value preference shocks gives the formal employment share

$$L_t^F = \Lambda \left(\frac{\log[(1 - \tau_t^F)w_t^F] - \log w_t^I - \kappa}{\mu} \right), \quad \Lambda(x) = \frac{1}{1 + e^{-x}}. \quad (9)$$

The parameter κ is the average nonpecuniary cost of formal employment. The scale μ controls heterogeneity. A positive μ prevents the economy from collapsing mechanically into a corner when after-tax wages change.

The formal payroll tax τ_t^F affects both the tax base and the allocation of labor. A higher tax lowers the net formal wage in (9). The resulting fall in L_t^F reduces the wage bill on which the tax is collected. The model therefore checks for an interior fiscal equilibrium rather than calculating the tax rate from a fixed base.

3.4 Population aging and fiscal closure

Let N_t^o denote people aged 65 and over and N_t^y people aged 15–64. Define the old-age dependency ratio

$$d_t \equiv \frac{N_t^o}{N_t^y}. \quad (10)$$

Each older adult requires a fixed real social-security outlay $\bar{g}_o$. After normalizing all aggregates by the working-age population, government spending is

$$G_t^{SS} = \bar{g}_o d_t. \quad (11)$$

Demography is exogenous in this version. The government does not choose G_t^{SS} independently of d_t and cannot reduce the per-capita commitment $\bar{g}_o$ along the experiment.

The government budget balances each period:

$$G_t^{SS} = \tau_t^F w_t^F L_t^F + T_t, \quad (12)$$

where T_t is a broad fiscal instrument represented as lump sum. Let $G_0^{SS} = \bar{g}_o d_0$ and $\Delta G_t^{SS} = G_t^{SS} - G_0^{SS}$. Required payroll revenue satisfies

$$\tau_t^F w_t^F L_t^F = \lambda_0 G_0^{SS} + \lambda_\Delta \Delta G_t^{SS}. \quad (13)$$

The remaining revenue is

$$T_t = (1 - \lambda_0)G_0^{SS} + (1 - \lambda_\Delta)\Delta G_t^{SS}. \quad (14)$$

All experiments share λ_0 and the same initial equilibrium. The broad-tax benchmark sets $\lambda_\Delta = 0$, mixed financing sets $\lambda_\Delta = 0.5$, and payroll-heavy financing sets $\lambda_\Delta = 0.75$. The broad instrument is a theoretical benchmark, not a claim that Colombia can collect nondistortionary taxes. Setting $\lambda_\Delta = 1$ is also evaluated as a feasibility check; under the present calibration it destroys the interior stationary formal-sector allocation.

3.5 Equilibrium

Given a demographic path $\{d_t\}$ and a marginal financing share, an equilibrium is a sequence

$$\{C_t, K_{t+1}, L_t^F, \tau_t^F, w_t^F, r_t, Y_t, G_t^{SS}\}_{t=0}^\infty$$

that satisfies household optimality (2), capital accumulation (3), factor prices (5)–(6), endogenous formality (9), the demographic spending rule (11), and the government budget (12)–(14).

In a stationary equilibrium,

$$r^* = \frac{1}{\beta} - 1 \quad (15)$$

and

$$\frac{K^*}{L^{F,*}} = \left[\frac{\alpha}{\beta^{-1} - 1 + \delta} \right]^{1/(1-\alpha)}. \quad (16)$$

Equations (9), (13), and (16) jointly determine formal employment, the payroll tax, and capital.

4 Calibration and experiment

Table 1 reports the initial calibration. The model targets 55.4 percent national labor informality, measured by [Departamento Administrativo Nacional de Estadística \(2025a\)](#). Public pension spending is set to 5.7 percent of GDP, as reported for Colombia by [Organisation for Economic Co-operation and Development \(2023\)](#). This figure fixes the scale of the fiscal outlay; it does not turn pension transfers into public consumption in national-accounting terms.

The demographic ratios are calculated from DANE’s July 2025 national population projections by single year of age. The numerator is population aged 65 and over and the denominator is population aged 15–64. The resulting ratio is 0.1502 in 2024 and 0.2728 in 2050 (Departamento Administrativo Nacional de Estadística, 2025b). The per-elderly outlay is calibrated from the initial identity

$$\bar{g}_o = \frac{G_0^{SS}}{d_{2024}}.$$

It is then held fixed.

Table 1: Initial calibration and demographic experiment

Parameter	Symbol	Value	Status or target
Capital share	α	0.33	Provisional standard value
Annual depreciation	δ	0.06	Provisional standard value
Discount factor	β	0.96	Provisional standard value
Risk aversion	σ	2.00	Provisional standard value
Informal productivity	A_I	0.60	Normalization
Choice dispersion	μ	0.30	Provisional
Informal employment	L^I	0.554	DANE target
Initial social-security outlays / GDP	G_0^{SS}/Y_0	0.057	OECD target
Old-age dependency ratio, 2024	d_{2024}	0.1502	DANE projection
Old-age dependency ratio, 2050	d_{2050}	0.2728	DANE projection
Outlay per older adult	$\bar{g}_o$	0.4285	Calibrated identity
Baseline payroll share	λ_0	0.50	Provisional closure
Implied increase / initial GDP	$\Delta G^{SS}/Y_0$	0.0465	Implied by aging

Notes: Aggregates are normalized by the working-age population. The calibration is illustrative. Parameters labeled “provisional” have not been estimated for this model. The formality cost κ is selected to match the informality target conditional on the remaining parameters.

For the numerical transition, the demographic comparison is represented as an unanticipated permanent shift:

$$d_t = \begin{cases} d_{2024}, & t = 0, \\ d_{2050}, & t \geq 1, \end{cases} \quad G_t^{SS} = \bar{g}_o d_t. \quad (17)$$

This experiment is a long-run comparative transition, not DANE’s literal annual path from 2024 to 2050. The step isolates the economy’s stable adjustment after the demographic burden has changed. A dated projection exercise should feed the full annual series into a nonstationary boundary-value solver.

5 Results

5.1 Stationary effects

Table 2: Population aging: stationary effects by financing rule

Financing of additional spending	ΔK (%)	ΔY (%)	ΔC (%)	Δ informality (p.p.)	$\Delta \tau_f$ (p.p.)
Broad-tax benchmark	0.00	0.00	-5.78	0.00	0.00
Mixed financing	-13.18	-6.17	-11.19	5.88	6.58
Payroll-heavy financing	-22.94	-10.75	-15.20	10.23	11.37

Notes: The old-age dependency ratio rises from 15.02 percent in 2024 to 27.28 percent in 2050, following DANE projections. Spending per older adult is fixed. Broad-tax financing places none of the additional outlay on the formal wage bill; mixed and payroll-heavy financing place 50 and 75 percent, respectively.

Table 2 separates the resource effect of aging from the formal-tax effect. Under the broad-tax benchmark, the formal employment wedge does not change. The stationary return condition fixes the formal capital–labor ratio, so capital, output, and informality remain at their initial values. The larger social-security claim is absorbed by private consumption, which falls 5.8 percent.

With mixed financing, half of the additional outlay is collected from the formal wage bill. The payroll tax rises 6.6 percentage points. Informality increases from 55.4 to 61.3 percent, capital falls 13.2 percent, output falls 6.2 percent, and private consumption falls 11.2 percent. Formal labor and capital are complements, so the contraction of the formal tax base also lowers the capital stock.

Payroll-heavy financing places 75 percent of the additional outlay on formal payroll. The payroll tax rises 11.4 percentage points and informality reaches 65.6 percent. Capital falls 22.9 percent, output 10.8 percent, and private consumption 15.2 percent. Requiring payroll to finance 100 percent of the aging-induced increase produces no interior stationary allocation. As the tax rate rises, formal employment and the wage bill contract too strongly to meet the revenue requirement. The model therefore signals a fiscal limit rather than reporting an artificial corner as an equilibrium.

5.2 Transition

Figure 2 reports the perfect-foresight transition implied by (17). Capital is predetermined when the higher dependency ratio is revealed. Consumption and fiscal instru-

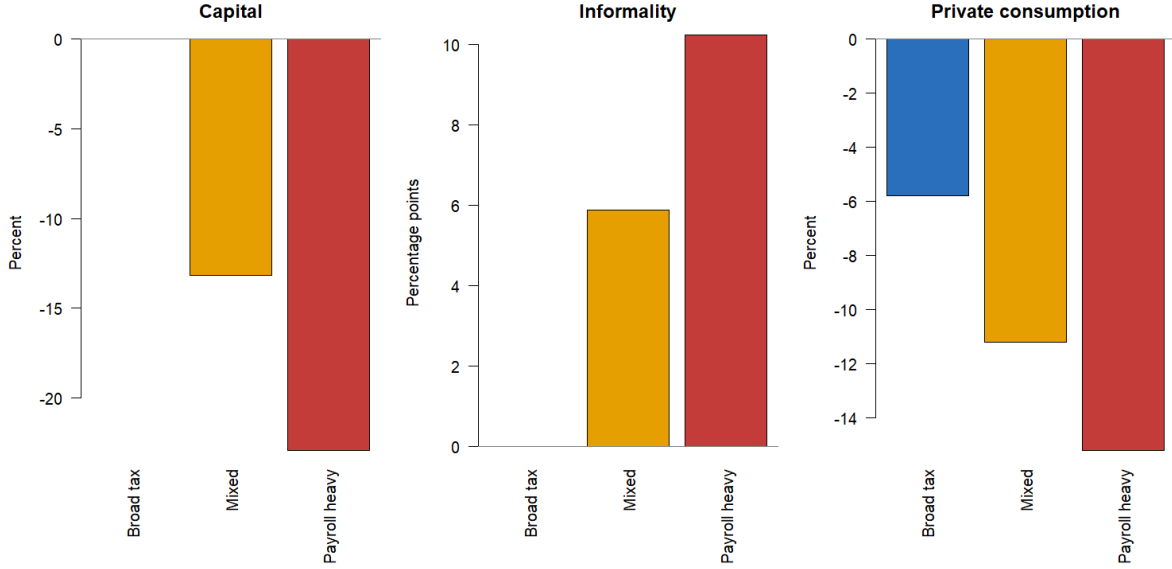


Figure 1: Stationary effects of aging by marginal financing instrument
Notes: Changes are measured relative to the common 2024 benchmark. The old-age dependency ratio is replaced by its DANE 2050 projection while spending per older adult remains fixed.

ments adjust immediately. Under broad-tax financing, private consumption absorbs the additional social outlay on impact and capital remains at its initial stationary value. Under mixed and payroll-heavy financing, the higher formal tax reduces formal employment and capital declines toward the lower stationary allocation.

The endogenous paths are monotone near the new steady state. The algorithm selects the stable branch of the Euler and resource equations; it does not smooth, average, or damp the computed series. Numerical residuals converge at the terminal allocation.

5.3 Social spending and welfare

The allocation is identical whether $\chi_G = 0$ or $\chi_G > 0$ in (1); only evaluated welfare changes. With $\chi_G = 0$, the permanent proportional increases in private consumption required to restore benchmark welfare are 6.1 percent under broad-tax financing, 9.5 percent under mixed financing, and 11.8 percent under payroll-heavy financing. With $\chi_G = 0.15$, those compensations become -1.9 , 1.3 , and 3.4 percent, respectively.

The negative number in the broad-tax case does not establish a welfare gain from Colombian pension spending. It reflects the assumed utility value of an aggregate that the model also treats as public absorption. A welfare analysis of pensions must instead value retirees' consumption and distinguish who pays from who receives.

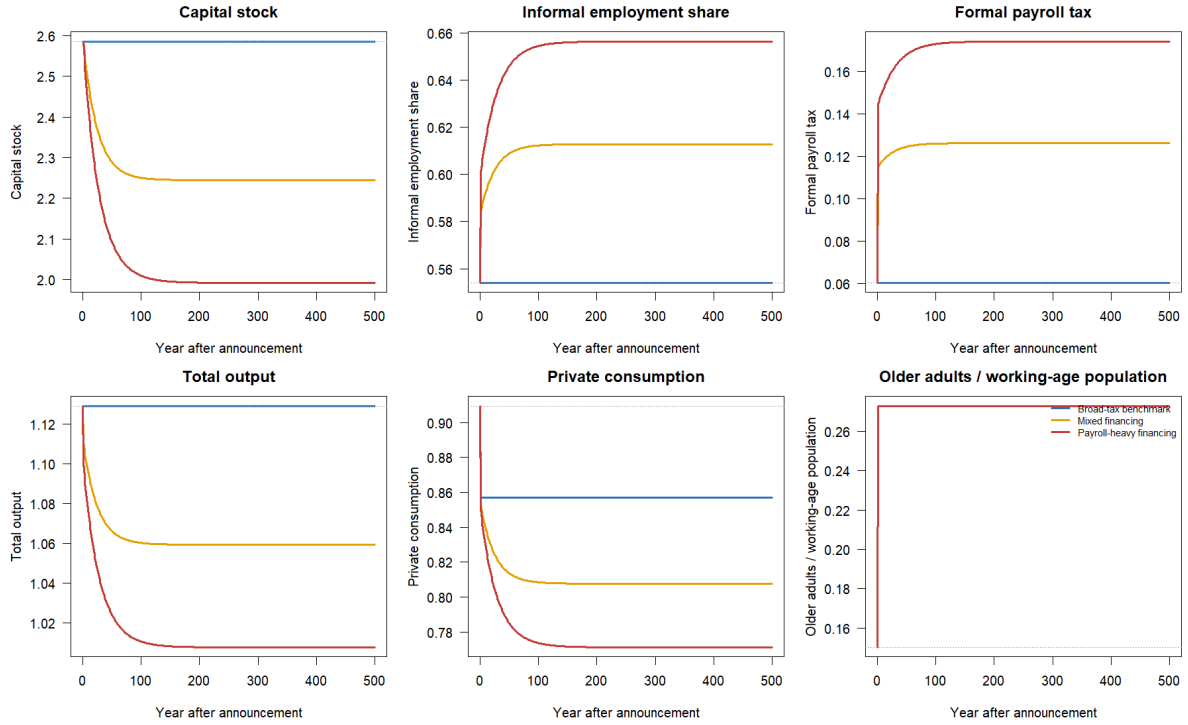


Figure 2: Transition after a permanent increase in the old-age dependency ratio
Notes: The gray dotted line is the common benchmark. The exogenous ratio moves from d_{2024} to d_{2050} at year 1. The long numerical horizon allows the stable root to approach the new stationary equilibrium. One period is one year; the horizontal axis is model time, not calendar time.

6 Extensions and limitations

The main limitation is the treatment of social-security outlays. Cash pensions are transfers; health and long-term care absorb real resources. Combining them in G_t^{SS} is useful for a first mechanism exercise but not for welfare or national-accounting decomposition. An overlapping-generations model should assign taxes to workers, benefits and health services to older households, and allow both groups to save.

Second, $\bar{g}_o$ is fixed. Actual pension benefits depend on contribution histories, replacement rules, indexation, and coverage. Health expenditure also varies sharply within the older population. A quantitative application should replace the scalar $\bar{g}_o$ with age-specific and program-specific schedules.

Third, the budget balances every year. Debt could smooth the demographic transition, but a permanent increase in d_t still requires a long-run primary adjustment. A debt extension should add an intertemporal government budget and a terminal fiscal rule rather than use debt as a residual.

Fourth, the broad-tax instrument is represented as lump sum. Observed consumption and income taxes have their own behavioral margins. The current comparison therefore identifies the additional distortion created by formal payroll financing, not an optimal tax mix.

Fifth, informal production uses labor only and formal and informal output are perfect substitutes. Allowing informal capital and aggregating the two goods with a finite elasticity of substitution could change the quantitative response. Those extensions should be assessed after separating pensions from real public services, because the latter distinction is more fundamental.

Finally, the demographic transition is imposed as a shift between two stationary ratios. A forecast should use the DANE annual path, endogenous labor force participation, mortality, migration, and cohort-specific retirement.

7 Conclusions

The fiscal experiment can be stated in one equation: $G_t^{SS} = \bar{g}_o d_t$. Holding social-security spending per older adult fixed, DANE's projected increase in Colombia's old-age dependency ratio raises the long-run revenue requirement by 4.65 percent of initial GDP in the illustrative calibration.

That demographic pressure does not have a unique effect on capital or informality.

Under the broad-tax benchmark, stationary capital and informality do not change, although private consumption falls. When half of the additional outlay is financed from formal payroll, capital falls 13.2 percent and informality rises 5.9 percentage points. With a 75 percent payroll share, capital falls 22.9 percent and informality rises 10.2 percentage points. Full marginal payroll financing has no interior stationary equilibrium.

The model's strongest conclusion is therefore about financing, not the exact size of the simulated effects. Aging raises the fiscal claim mechanically; placing that claim on a tax base that contracts with its own rate amplifies the adjustment through informality and capital accumulation.

The next step should be structural rather than cosmetic: separate pension transfers from health and care consumption, introduce working and retired households, feed the annual DANE demographic path into the transition, and replace the broad benchmark with observed Colombian taxes and a debt rule. Those additions would turn the present mechanism exercise into a model of intergenerational social security.

A Numerical algorithm

The code is written in R and uses only base packages. The stationary and transition calculations proceed as follows.

1. Given the exogenous populations, compute $d_t = N_t^o/N_t^y$ and $G_t^{SS} = \bar{g}_o d_t$.
2. Given (K_t, G_t^{SS}) and required payroll revenue, solve (9) jointly with (13). The algorithm searches all admissible roots and selects the root connected to the preceding allocation.
3. Compute formal and informal output, factor prices, tax revenue, and the government budget residual.
4. In a stationary equilibrium, impose (15) and solve the remaining scalar equation for formal employment. Compute consumption from

$$C^* = Y^* - G^{SS,*} - \delta K^*.$$

5. For the transition, combine (2) and (3) into a two-dimensional map for (K_t, C_t) . Capital at the announcement date is inherited from the benchmark.

6. Evaluate the Jacobian of that map at the final stationary equilibrium. The calibration has one stable and one unstable root. Select the eigenvector associated with the stable root.
7. Start close to the final equilibrium on the stable eigenvector and apply the inverse Euler and resource maps. Continue backward until the path reaches the inherited capital stock. A scalar root selects the terminal displacement that reproduces initial capital.
8. Reverse the resulting sequence and append the final stationary allocation after the terminal discrepancy is below the numerical tolerance. The reported tests verify the household Euler equation, the resource constraint, the government budget, the identity $G_t^{SS} = \bar{g}_o d_t$, the benchmark targets, and terminal convergence.

The stable-path method avoids the numerical instability of forward shooting. It does not smooth the endogenous variables after they are computed. The transition paths are generated by the model equations themselves.

B Alternative preference specifications

The additive specification in (1) can be replaced by a nonseparable private–public consumption aggregator:

$$X_t = \left[\omega^{1/\eta} C_t^{(\eta-1)/\eta} + (1 - \omega)^{1/\eta} (G_t^{SS})^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}. \quad (18)$$

Utility would then be $u(X_t)$. Social-security services change the marginal utility of private consumption and can affect saving during a transition. For cash pensions, however, utility should depend on recipients’ private consumption rather than on aggregate outlays. The CES model is economically different from assigning a positive value to χ_G in (1); it should be calibrated as a separate experiment.

References

- Barro, R. J. (1990). Government spending in a simple model of endogenous growth. *Journal of Political Economy*, 98(5):S103–S125.

- Baxter, M. and King, R. G. (1993). Fiscal policy in general equilibrium. *American Economic Review*, 83(3):315–334.
- Departamento Administrativo Nacional de Estadística (2025a). Empleo informal y seguridad social: agosto–octubre 2025. Technical report, DANE, Bogotá.
- Departamento Administrativo Nacional de Estadística (2025b). Serie nacional de población por área, sexo y edad, 2018–2070. Proyecciones de Población y Estudios Demográficos. Updated 18 July 2025.
- Galí, J., López-Salido, J. D., and Vallés, J. (2007). Understanding the effects of government spending on consumption. *Journal of the European Economic Association*, 5(1):227–270.
- Granda, C. and Morales, L. F. (2023). Tax policy, informality, and real wage rigidities. Borradores de Economía 1245, Banco de la República.
- Herrera-Rojas, A. N., López-Valenzuela, D. C., Ospina-Tejeiro, J. J., and Bejarano-Rojas, J. A. (2025). Modelo fiscal colombiano para la evaluación económica. Borradores de Economía 1324, Banco de la República.
- La Porta, R. and Shleifer, A. (2014). Informality and development. *Journal of Economic Perspectives*, 28(3):109–126.
- Levy, S. (2008). *Good Intentions, Bad Outcomes: Social Policy, Informality, and Economic Growth in Mexico*. Brookings Institution Press, Washington, DC.
- Mejía, D. and Posada, C. E. (2007). Informalidad: teoría e implicaciones de política. Borradores de Economía 455, Banco de la República.
- Organisation for Economic Co-operation and Development (2023). Pensions at a glance 2023: Country profile—colombia. OECD Publishing.
- Osorio-Copete, L. M. (2016). Reforma tributaria e informalidad laboral en colombia: un análisis de equilibrio general dinámico y estocástico. *Ensayos sobre Política Económica*, 34(80):126–145.
- Ospina-Tejeiro, J. J., Ramos-Forero, J. E., López-Valenzuela, D. C., Hernández-Turca, Y., and Herrera-Pinto, N. V. (2024). El sistema de pensiones en colombia: perspectivas y riesgos fiscales con base en las normas vigentes. Borradores de Economía 1271, Banco de la República.

- Rincón, H., Rodríguez, N., Toro, J., and Téllez, S. (2017). Fisco: modelo fiscal para colombia. *Ensayos sobre Política Económica*, 35(83):161–187.
- Ulyssea, G. (2018). Firms, informality, and development: Theory and evidence from brazil. *American Economic Review*, 108(8):2015–2047.